

The main idea is to keep the required storage conditions stable while allowing the system to be monitored and controlled wirelessly through a website.

Conditions to maintain: approximately 1–4°C temperature and 90–95% relative humidity, based on the pre-research assignment. The system should also support pre-cooling and continuous monitoring so that the cold chain is not broken after harvesting.

1. Main Controller and Communication

Component	use
ESP32	Main controller. Reads the sensors, runs the local control logic and connects the system to the website through Wi-Fi.
Wi-Fi router/access point	Provides wireless network connectivity between the controller and the website.
MicroSD card module	Provides local backup storage for sensor readings and events.
RTC module (DS3231)	Keeps accurate time for data logging and event records.
5 V/12 V power supplies and DC-DC converters	Provide the required regulated voltages for the electronics.

2. Temperature and Humidity Monitoring

Component	use
DS18B20 waterproof temperature sensors (2–3)	Measure temperature at different points inside the chamber so temperature differences can be detected.
SHT31/SHT35 temperature and humidity sensors (1–2)	Monitor both temperature and relative humidity. These are recommended for the main environmental readings.
Magnetic reed switch	Detects whether the cold-room door is open or closed.
Optional current sensor (ACS712/ACS758)	Monitors the electrical current used by major loads such as the compressor.

3. Cooling System

Component	use
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Compressor	Provides the refrigeration needed to bring the chamber down to the required temperature.
Condenser	Rejects heat from the refrigeration system.
Evaporator/cooling coil	Removes heat from inside the cold chamber.
Expansion valve or capillary tube	Controls refrigerant expansion in the refrigeration cycle.
Condenser fan	Helps remove heat from the condenser.
Evaporator/circulation fans	Circulate cooled air around the chamber.
Compressor contactor/relay	Allows the controller to switch the compressor safely.
Component	use
Thermal overload/safety thermostat	Provides independent protection for the refrigeration equipment.

4. Humidity Control

Component	use
Ultrasonic humidifier	Raises humidity when it falls below the required range.
Water reservoir	Stores water for the humidifier.
Water-level sensor	Detects low water and prevents the humidifier from running dry.
Dehumidifier or controlled exhaust/ventilation	Reduces excessive humidity when it rises above the set range.
Solenoid valve (optional)	Can be used if automatic filling of the humidifier reservoir is required.
Drainage tray/pipe	Handles condensate and excess water.

5. Air Circulation and Ventilation

Component	use
DC/AC circulation fans	Help maintain a more uniform temperature and humidity throughout the storage chamber.
Controlled exhaust fan	Provides controlled air exchange when required.

Motorized ventilation damper and actuator	Controls the amount of outside air entering the chamber.
Air ducts/vents	Direct air movement around the storage area.

6. Website and Remote Management

Component	use
Web dashboard	Shows live temperature, humidity, door status, actuator status and system condition.
Database	Stores historical readings for analysis.
MQTT or HTTP communication	Provides communication between the ESP32 and the web application.
REST API	Links the controller data with the website where required.
User authentication	Restricts control of the system to authorised users.
Graphs and alarms	Show trends and warn the user when temperature or humidity move outside the set range.

7. Optional Product Monitoring

Component	use
Camera module	Can periodically capture images of the stored crops for visual monitoring.
Raspberry Pi 5 (optional)	Can be added if image processing or AI-based quality/disease detection is required.
LED lighting	Provides consistent illumination for camera images.

8. Safety and Protection

Component	use
MCB/circuit breakers	Protect the electrical installation from overcurrent.
Fuses	Provide additional protection for individual circuits.
Emergency-stop button	Allows the system to be shut down quickly when necessary.
Buzzer and warning LEDs	Give local warnings for abnormal conditions.
High/low temperature alarms	Warn when temperature leaves the acceptable range.
High/low humidity alarms	Warn when humidity leaves the acceptable range.

Door-open alarm	Alerts the operator if the door remains open for too long.
Proper earthing and electrical isolation	Protects users and low-voltage electronics, especially where AC refrigeration equipment is used.

9. Cold-Room Structure

Component	use
Insulated cold-room/chamber panels	Reduce heat entering the chamber and improve refrigeration efficiency.
Insulated door and door gasket	Minimise heat leakage when the door is closed.
Food-safe internal surfaces	Provide a suitable storage environment.
Sensor mounting points and cable glands	Allow sensors and wiring to be installed neatly while maintaining insulation.

10. Power Supply (if an independent system is required)

Component	use
Solar panels	Generate electrical energy where solar power is desired.
Solar charge controller	Controls battery charging.
12/24 V battery	Stores energy for operation when solar power is unavailable.
Inverter (if AC refrigeration is used)	Converts battery DC to AC for suitable refrigeration equipment.
DC distribution, fuses and battery protection	Distribute and protect the power system.

Suggested Overall Arrangement

Sensors → ESP32 → Local control logic → Cooling/Humidity/Fans → Wi-Fi → Website

The website would allow the user to view live conditions, see historical graphs, change setpoints and switch to manual control when needed. I would keep the main control logic on the ESP32 so the cold chain can continue even if the internet connection is temporarily unavailable. The website should therefore be treated as the remote monitoring and management layer, rather than the only controller.

Important: The refrigeration equipment should have its own appropriate electrical and thermal protection. The exact compressor, evaporator, condenser and power

ratings should be selected after deciding the chamber size, crop load, ambient temperature and required pull-down time.